# Alcohols

**Exam code:** 0620 & 0971

Download PDF

**Written by:** Alexandra Brennan
**Reviewed by:** Stewart Hird
Updated on 30 September 2025

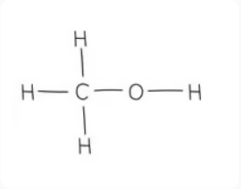
SaveMyExams

Did this video help you? Yes No

## Alcohols

- All alcohols contain the hydroxyl (**–OH**) functional group which is the part of alcohol molecules that is responsible for their characteristic reactions
- Alcohols are a homologous series of compounds that have the general formula $\mathbf{C_nH_{2n+1}OH}$
- They differ by one $-CH_2$ in the molecular formulae from one member to the next

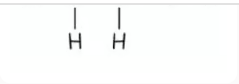
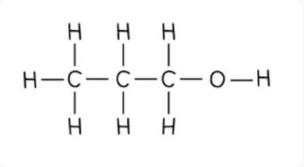
**Table showing the first three alcohols**

| Name | Formula | Displayed formula |
|---|---|---|
| Methanol | $CH_3OH$ |  |

**Stuck on something?**

Ask questions and get help with exactly what you're revising.

Got it

| | | |
|---|---|---|
| | |  |
| Propanol | $C_3H_7OH$ |  |

- Ethanol ($C_2H_5OH$) is one of the most important alcohols
  - Ethanol can also be represented by its structural formula $CH_3CH_2OH$
- It is the type of alcohol found in **alcoholic drinks** such as wine and beer
- It is also used as **fuel** for cars and as a **solvent**
- Ethanol will undergo complete combustion and burn in excess oxygen to produce carbon dioxide and water

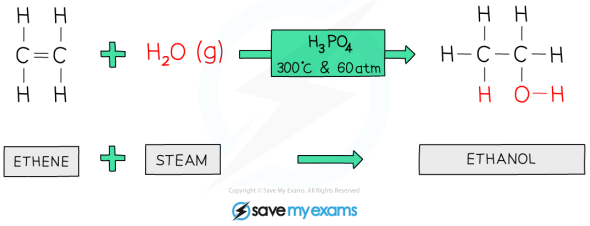
$$C_2H_5OH\,(l) + 3O_2\,(g) \rightarrow 2CO_2\,(g) + 3H_2O\,(l)$$

# The manufacture of ethanol

- There are two methods used to manufacture ethanol:
  - The **hydration** of **ethene** with steam
  - The **fermentation** of glucose
- Both methods have advantages and disadvantages which are considered

## Hydration of ethene

- A mixture of ethene and steam is passed over a hot catalyst of **phosphoric acid** at a temperature of approximately **300 °C**
- The pressure used is **60 atmospheres (6000kPa)**
- The gaseous ethanol is then condensed into a liquid for use




***A water molecule adds across the C=C in the hydration of ethene to produce ethanol***

## Fermentation of glucose

- Sugar or starch is dissolved in water and yeast is added

starch or sugar to glucose

- If the temperature is too **low** the reaction rate will be too slow and if it is too **high** the enzymes will become **denatured**
- The yeast respire anaerobically using the glucose to form ethanol and carbon dioxide:

$$C_6H_{12}O_6 \rightarrow 2CO_2 + 2C_2H_5OH$$

- The yeast are killed off once the concentration of alcohol reaches around 15%, so the reaction vessel is emptied and the process is started again
- Ethanol production by fermentation is therefore a **batch** process

### Examiner Tips and Tricks

Make sure you learn the conditions for both hydration and fermentation.

# Comparing methods of ethanol production

**Extended tier only**

| | **Hydration of ethene** | **Fermentation** |
|---|---|---|
| **Equipment** | complex set up required | simple equipment needed |
| **Raw materials** | uses non-renewable resources (crude oil) | uses renewable resources (sugar cane) |
| **Type of process** | continuous process- a steam of reactants is constantly passed over a catalyst | batch process- everything is mixed together in a reaction vessel and left for several days and the process repeated again |
| **Rate of reaction** | fast | very slow (several days) |
| **Quality of product** | produces pure ethanol | produces a dilute solution requiring further processing |

| | | |
|---|---|---|
| | formed from the burning of fossil fuels to maintain high temperatures | |
| **Reaction conditions** | high temperatures and pressures required increasing the energy input and cost | low temperatures required |

Teacher only

## New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder

Test yourself

Flashcards

Was this revision note helpful? Yes No

Previous

Next

Course

Study tools

All topics

**Resources**

Learning Hub

Scholarship

Join

Past Papers

Solution Banks

**Dashboard**

Overview

Brand ambassador

Profile

Subscription

Settings

About us
Exam Specificity
Content Quality
Promotions
Jobs
Terms
Privacy
Cookie Policy
Help and Support

Biology
Chemistry
Physics
Maths
Geography
English Literature
Psychology
All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.
Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.